

Simulate Your Circuits Using Qucs

In this article we explore the features and functions of Quite Universal Circuit Simulator (Qucs) and go through a step-by-step analysis. Qucs is available for Windows, Mac and Ubuntu (open source) in the DVD accompanying this month's issue of EFY Plus

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An open source electronics circuit simulator software released under GPL, Quite Universal Circuit Simulator (Qucs) provides the user with capabilities to set up a circuit with a graphical user interface. It can simulate circuit behaviour, and supports pure digital simulations using VHDL and/or Verilog languages.

Qucs is much simpler to use and handle than other circuit simulators like gEDA or PSpice. It also supports a growing list of analogue and digital components as well as Spice sub-circuits, which makes it easier to work with.

What is Qucs all about

Qucs is an integrated circuit simulator that enables you to set up a circuit with its intuitive graphical user interface (GUI) and simulate the small-signal, large-signal and noise behaviour of the circuit. Post simulation, you can view the simulation results on a presentation page or window.

Qucsator and GUI

The GUI is based on Qt by Digia. The software aims to support most kinds of circuit simulation like DC, AC, S-parameter, harmonic balance analysis and noise analysis to name a few. The

Top three functions of Qucs

1. Schematic capture
2. Analogue and digital simulations
3. Data visualisation

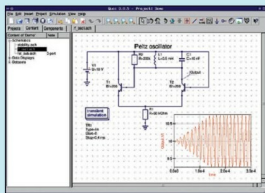


Fig. 1: Combined schematic and data visualisation

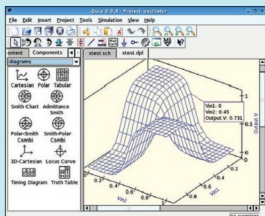


Fig. 2: Data display with 3D diagram

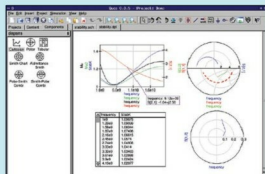


Fig. 3: Data display with different kinds of data representations (diagrams) and data markers

Qucs GUI is well advanced and allows setting up schematics and presenting simulation results in various types of diagrams. Apart from representation in DC, AC, S-parameter, noise and transient analysis, mathematical equations and use of a sub-circuit hierarchy (with parameterised sub-circuits) are also available. Qucs can also import existing Spice models for use in your simulations.

Qucsator, the simulation back-end, is a command-line circuit simulator. It takes a network list in a certain format as input and outputs a Qucs dataset. It has been programmed for usage in the Qucs project but may also be used by other applications.

Qucs comes with a huge array of components and models including HICUM, BSIM2, BSIM3 and BSIM6 to list a few. It also provides semiconductor-based components and models such as PMOSFETs, MOSFETs, op-amps, diodes and many more.

While some examples have been included in this story to demonstrate some of the abilities of Qucs, many more are available within the program in the DVD.

Features and their functions

Qucs consists of several standalone programs interacting